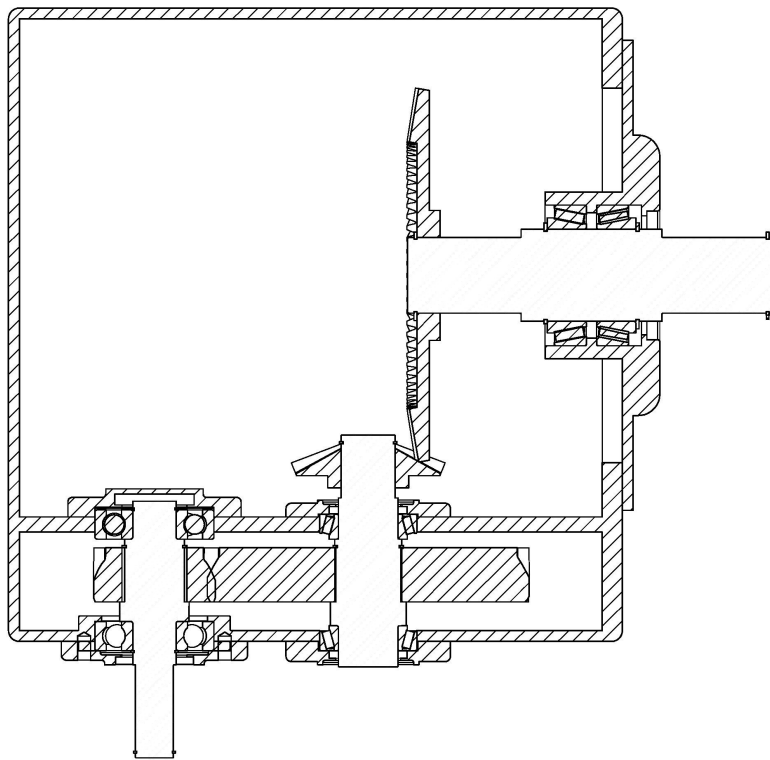


Two-Stage Gearbox Design

Mechanical analysis, component selection and SolidWorks assembly



11 kW

DESIGN POWER

7.23 : 1

INTERNAL RATIO

3 shafts

Ali Sakhaei | Mohammadreza Sotude

Sharif University of Technology | Academic team project | 2025-2026

Design Basis and Transmission Layout

The project develops a complete two-stage gearbox from analytical sizing to a detailed SolidWorks assembly. The transmission uses a helical parallel-axis stage followed by a straight bevel stage to redirect the output axis.

11 kW

MOTOR POWER

1440 rpm

MOTOR SPEED

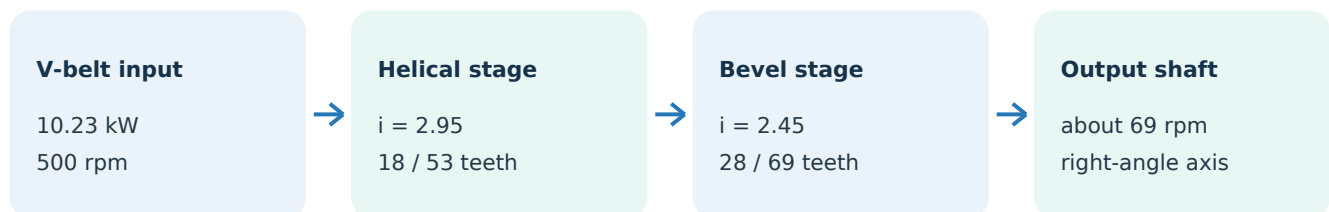
500 rpm

SHAFT 1 SPEED

10 years

DESIGN LIFE

POWER FLOW



CORE ASSUMPTIONS AND CHOICES

- Effective power after the V-belt is 10.23 kW, giving 195.4 N m torque at the first gearbox shaft.
- Gear material: quenched and tempered 34CrMo4, with design checks based on bending and contact stress.
- Shaft material: CK45 steel; stepped diameters were sized under combined bending and torsion.
- A 99% reliability target and practical standard sizes were used where specified in the original calculations.

CALCULATION FRAMEWORK

Torque from power and speed

$$T = \frac{9550P}{n}$$

Tangential tooth load

$$F_t = \frac{2T}{d_p}$$

Stage ratio

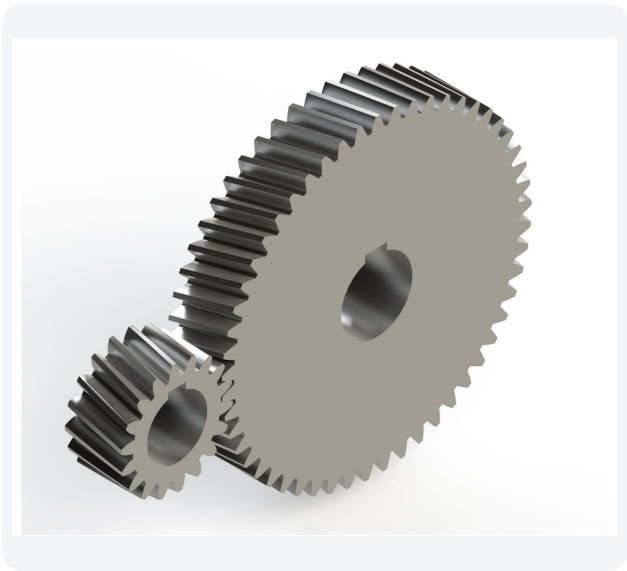
$$i = \frac{z_2}{z_1}$$

Basic bearing life

$$L_{10} = \left(\frac{C}{P}\right)^p$$

Helical and Bevel Gear Stages

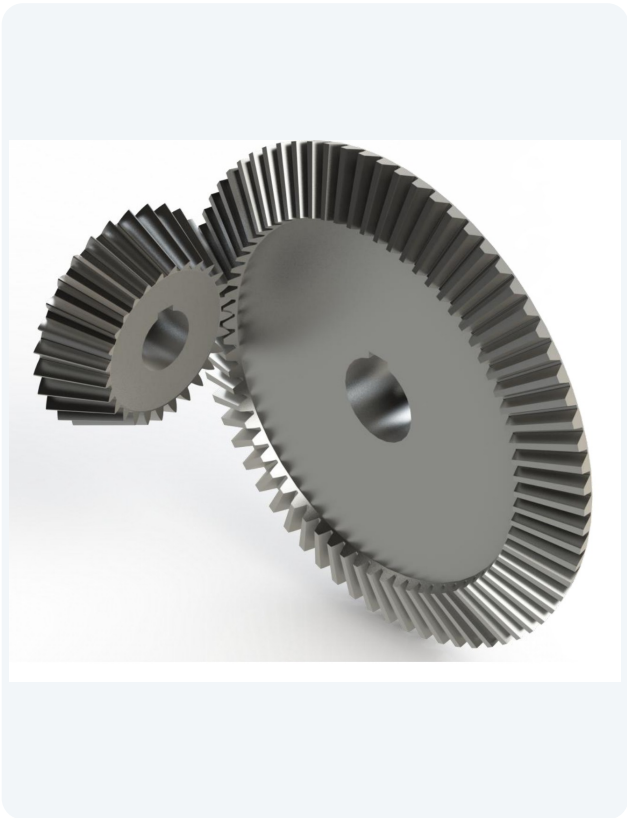
STAGE 1 - SELECTED HELICAL PAIR



Parameter	Selected value
Speed ratio	2.95 : 1
Tooth count	18 / 53
Normal module	5 mm
Helix angle	13 deg
Pitch diameters	92.4 / 272 mm
Face width	about 51 mm

Both spur and helical alternatives were evaluated. The helical pair was carried forward for the detailed gearbox layout. The selected geometry is based on standard module sizing after bending and surface durability checks.

STAGE 2 - STRAIGHT BEVEL PAIR



Parameter	Selected value
Speed ratio	2.45 : 1
Tooth count	28 / 69
Module	6 mm
Pitch diameters	168 / 414 mm
Face width	58 mm
Pitch angles	22.1 / 67.9 deg

Verified operating point

Power entering stage: 9.92 kW
Shaft speed: about 170 rpm
Transmitted torque: 557.2 N m

Shaft Design and Bearing Selection

Three stepped shafts were designed around the calculated gear forces. Combined bending and torsion were checked using fatigue stress concentration factors, then the diameters were rounded to practical bearing seats and keyway sizes.



SHAFT 1



SHAFT 2



SHAFT 3

BEARING ARRANGEMENT

Shaft	Bearing	Arrangement	Design note
1	6408 deep-groove	Locating / non-locating	10-year duty basis
2	32011 X/Q tapered	Face-to-face pair	40.8 Mrev calculated life
3	33017/Q tapered	Back-to-back pair	167.8 Mrev calculated life

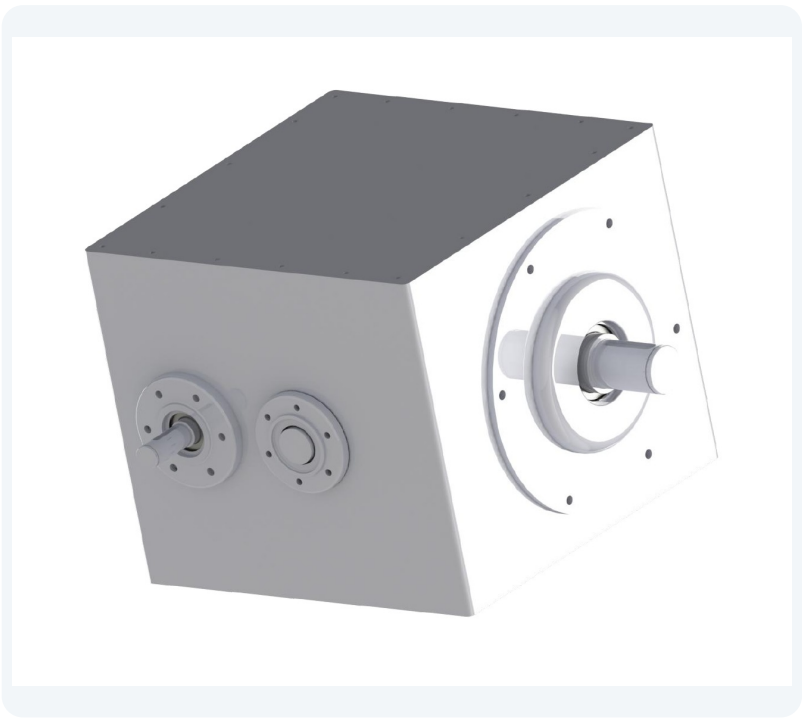
KEY DESIGN CHECKS

- Gear forces were resolved into tangential, radial and axial components before reaction and moment calculations.
- Critical shaft sections were evaluated at shoulders, bearing seats, gear seats and keyways.
- Keys and retaining rings provide torque transfer and axial retention at the specified interfaces.
- Bearing arrangements were selected to manage radial load, thrust load, shaft location and assembly access.

Design equation used for shaft sizing

$$d = \left[\frac{32n}{\pi} \sqrt{\left(\frac{K_f M_a}{S_e} \right)^2 + \left(\frac{K_{fs} T_m}{S_y} \right)^2} \right]^{1/3}$$

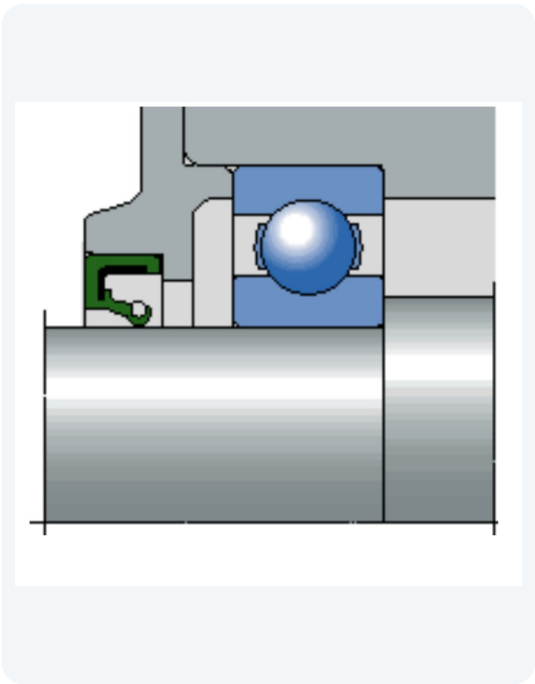
Housing, Fasteners and Sealing



HOUSING DESIGN

- Steel plate construction was checked against bearing reaction loads.
- Calculated local minima were increased to practical 10-12 mm plate selections.
- Removable covers, bearing flanges and a dedicated output-shaft housing support assembly and service access.
- Fastener sizes include M10, M12, M14 and M16 selections across the casing and support interfaces.

RADIAL SHAFT SEALING



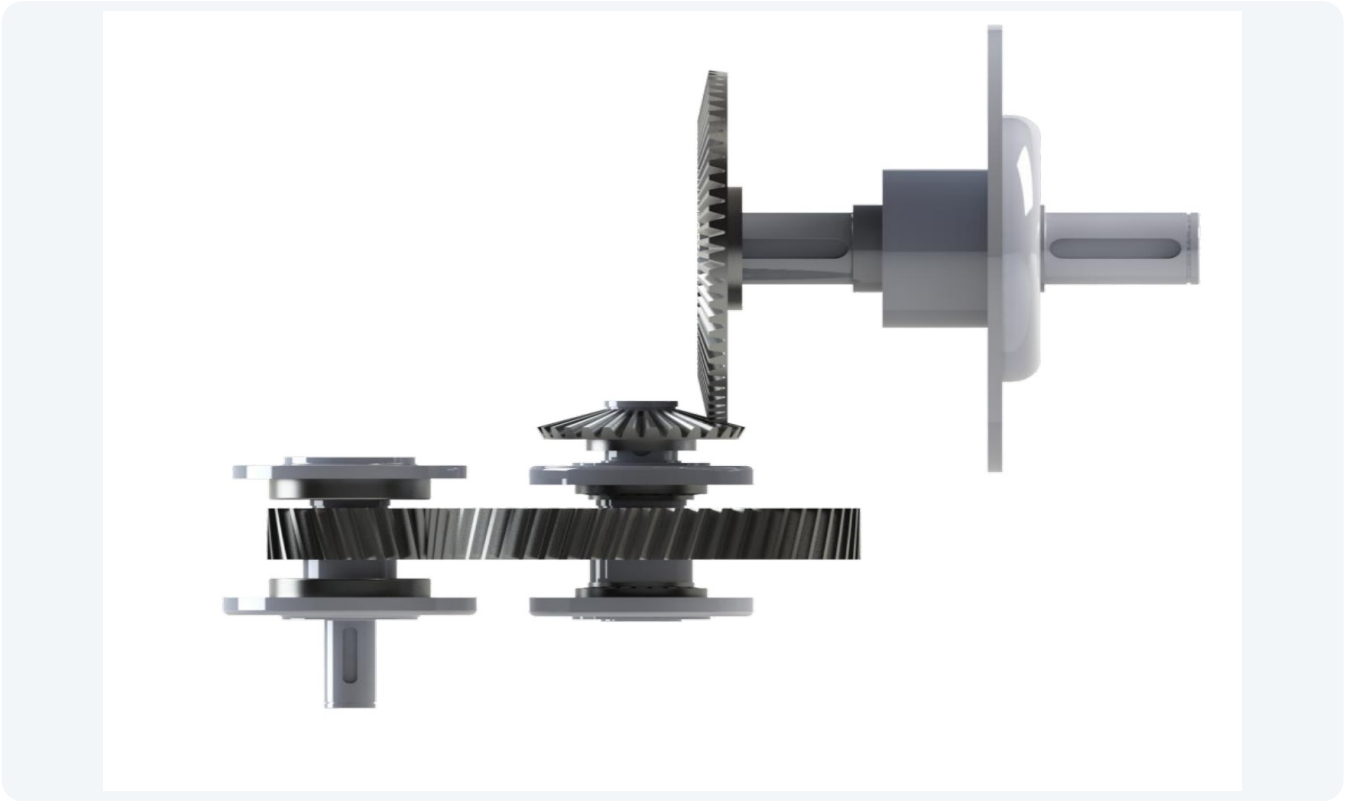
Location	Selected seal
Behind B1	Type A 40-72-7
Behind B3	Type A 55-80-8
Behind B6	Type A 85-120-12

Type A radial lip seals were selected to limit lubricant leakage and contaminant ingress. Seal seats were integrated into the bearing flanges and the third-shaft housing.

Manufacturing intent

Use standard plate thicknesses, standard rolling bearings and catalog seals to keep the design serviceable.

Assembly Architecture and Deliverables



ASSEMBLY LOGIC

- Shaft 1 carries the helical pinion and is supported by the locating/non-locating ball-bearing pair.
- Shaft 2 carries the helical gear and bevel pinion, with a face-to-face tapered-roller arrangement.
- Shaft 3 carries the bevel gear and output element, supported by a back-to-back tapered-roller pair.

REPOSITORY-READY DELIVERABLES

Deliverable	Contents
SolidWorks source files	Parts, subassemblies, main assembly and drawings
Engineering drawings	Gears, shafts, flanges, housing and sectional views
Persian calculation report	Complete academic report with detailed calculations
English portfolio report	Concise design summary and selected results

Engineering Outcomes

FINAL CONFIGURATION

Item	Final selection
Architecture	Two-stage, three-shaft, right-angle gearbox
First stage	Helical pair, $i = 2.95$, module 5 mm
Second stage	Straight bevel pair, $i = 2.45$, module 6 mm
Internal reduction	Approximately 7.23 : 1
Gear material	34CrMo4, quenched and tempered
Shaft material	CK45 steel
CAD platform	SolidWorks

WHAT THE PROJECT DEMONSTRATES

- Translation of power, speed and service-life requirements into a complete transmission layout.
- Gear sizing with bending and surface durability checks, including comparison of spur and helical alternatives.
- Integrated design of shafts, bearings, keys, retaining rings, seals, housing plates and fasteners.
- CAD assembly planning supported by manufacturing drawings, an exploded view and a bill of materials.

REPORT SCOPE

This English edition is a concise portfolio summary based on the complete Persian calculation report. It preserves the selected dimensions and component choices while omitting student numbers and personal contact details.

PROJECT CREDIT

Academic team project by Ali Sakhaei and Mohammadreza Sotude.

Repository: github.com/alisakhaei/gearbox-design